

The adrenal glands are divided into two anatomically distinct regions: the adrenal medulla and the adrenal cortex. The adrenal medulla is a collection of irregularly shaped cells located in the interior of each adrenal gland. In some instances, increased serum concentrations of corticosteroid hormones can stimulate medullary cells to synthesize and secrete the catecholamines norepinephrine and epinephrine. Norepinephrine and epinephrine are water-soluble compounds that can function as both hormones and as neurotransmitters, and their secretion into the bloodstream is stimulated by acetylcholine (ACh)-mediated signaling in the sympathetic component of the autonomic nervous system. Figure 1 shows the enzyme-mediated synthesis of norepinephrine and epinephrine.

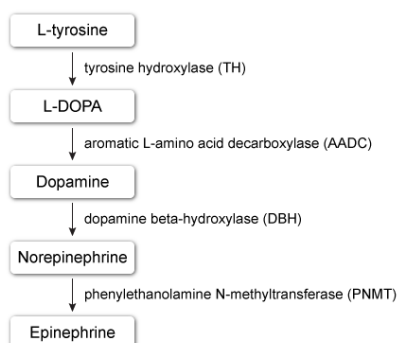


Figure 1

The adrenal cortex is the outer portion of the adrenal gland and is composed of cells that synthesize and secrete lipophilic hormones derived from cholesterol. Hormones produced by the adrenal cortex include the principal glucocorticoid hormone (cortisol) and the principal mineralocorticoid hormone (aldosterone).

The hypothalamus and pituitary gland regulate and control the endocrine functions of the adrenal cortex (Figure 2). Secretion of corticotropin-releasing hormone (CRH) from the hypothalamus promotes the synthesis and release of adrenocorticotropic hormone (ACTH) by the pituitary gland. ACTH acts on adrenal cortex cells to promote the growth of cortical tissue and to increase the synthesis and secretion of glucocorticoids. In addition to its body-wide functions as the principal stress hormone, cortisol also regulates adrenal activity by inhibiting CRH release from the hypothalamus and ACTH release from the pituitary.

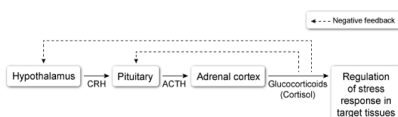


Figure 2

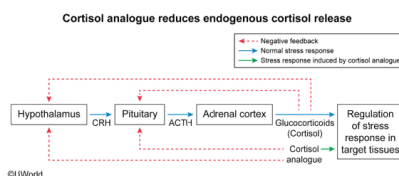
A woman with low serum cortisol due to partial adrenal dysfunction is treated with high doses of a synthetic cortisol analogue. According to Figure 2, treatment with a cortisol analogue will most likely cause this patient's serum level of endogenous cortisol to:

- ☐ A. increase only. [4%]
- ☒ B. decrease only. [47%]
- ☐ C. remain the same. [4%]
- ☐ D. increase at first, but then decrease over time. [43%]

Correct

46% answered correctly

Explanation:



Homeostasis refers to the ability of an organism to maintain a relatively stable physiological environment in response to external or internal changes. One mechanism of homeostasis is **negative feedback**, the process by which the product of a metabolic or signaling pathway *inhibits* one or more of the steps that promote the synthesis or secretion of that product. For example, as the serum level of a given hormone rises, that hormone inhibits the signaling pathway that promotes the synthesis and secretion of itself.

Figure 2 shows the negative feedback pathway that influences the synthesis and secretion of endogenous (ie, produced in the body) **cortisol**. The **hypothalamus** secretes CRH, which then induces the **anterior pituitary gland** to release **ACTH**. In turn, ACTH stimulates the adrenal glands to synthesize and secrete the glucocorticoid hormone cortisol. In addition to mediating the stress response, cortisol also inhibits CRH release from the hypothalamus and ACTH release from the pituitary gland. By *inhibiting* the secretion of CRH and ACTH, secreted cortisol *increases* negative feedback such that the adrenal cortex receives *less* stimulatory input and the synthesis and secretion of additional cortisol *decrease*.

Synthetic **cortisol analogues** have a molecular structure that resembles that of endogenous cortisol, and therefore are able to mimic the effect of cortisol by binding hypothalamic and pituitary cortisol receptors. Although the *analogue* concentration is determined by the amount administered, interaction of the analogue with cortisol receptors will induce *negative feedback* that *reduces* the synthesis and secretion of *endogenous* cortisol by the adrenal glands. Therefore, after **treatment with a synthetic cortisol analogue**, the patient's **serum level of endogenous cortisol would decrease** due to negative feedback.

(Choices A, C, and D) Administration of a synthetic cortisol analogue increases the activation of the cortisol receptors, mimicking the physiologic effects of cortisol *without* increasing the serum level of endogenous cortisol. This analogue will also increase negative feedback, *diminishing* the synthesis of endogenous cortisol and ultimately decreasing its serum concentration.

Educational objective:

Negative feedback is the process by which the product of a metabolic or signaling pathway inhibits one or more of the steps that promote the synthesis or secretion of that product.

Time spent: 0 sec
Subject:
Biology

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Endocrine and Nervous Systems